



⇒ To build a system that estimates what electricity prices should be based on real world factors, then use that to help make trading decisions & manage risk

⇒ Unlike stock, power is affected by things like weather, consumption.

⇒ What I am trying to do know whether supply exceed demand or not using the factors to know the price.

⇒ Load → how much electricity people are using, measured in megawatts
higher load → higher prices.

⇒ Wind → high wind → low prices

high load + low wind → Very high price

low load + high wind → low price

high load + high wind → Mixed

low load + low wind → Moderate

⇒ predict tomorrow's hourly day-ahead electricity prices, then use those hourly forecasts to estimate next week's or next month's average price.

Target → Next-day hourly day-ahead price DE-LU, then average them to get base, peak, Next-week avg & Next-month avg.

Justification — I Choose next-day hourly day ahead prices because day-ahead prices are the most directly linked to hourly fundamentals such as load & wind generation. Forecasting hourly prices also allows the model output to be aggregated into base / peak blocks & next week or next-month expected averages making it relevant for prompt curve positioning

We need one baseline model the improved model.

We Can't use future data to predict the past
↳ time-series appropriate validation.

Models must work on unseen future market
Use MAE (mean absolute error) or RMSE to
know how good our prediction is.

⇒ Explain how our model predicts day-ahead
prices, but how would an actual trader
use that forecast to trade the curve.

⇒ how do we change the day-ahead forecast
into a broader trading signal.

⇒ Need to know the market risk to know
when the model could be wrong.

⇒ use LLM → not manually

